Static Displays

In this document, I'm going to create some static displays so I get some practice creating summaries that will eventually be in my Shiny app.

Prepare for your App

##Cleaning things up a bit

The first thing I'm going to do is create a tibble out of the data in my spreadsheet.

```
# Load the tidyverse, ggplot and save the data to a tibble with a short
library(tidyverse)
```

```
Warning: package 'readr' was built under R version 4.5.3
```

```
Warning: package 'dplyr' was built under R version 4.5.3
```

```
Warning: package 'lubridate' was built under R version 4.5.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.5     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```r
library(ggplot2)
#Create student data tibble

student_data<-read_csv("data.csv",locale=locale(encoding="latin1"))
```

```
Rows: 186 Columns: 28
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (11): Start time, Completion time, Email, What kind of pet do you have? ...
dbl (15): ID, How many pets do you have?, How far (in miles) did you commute...
lgl  (2): Name, Last modified time

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
student_data
```

```
# A tibble: 186 x 28
      ID `Start time`    `Completion time` Email     Name  `Last modified time`
   <dbl> <chr>           <chr>             <chr>     <lgl> <lgl>
 1     3 8/27/2024 10:59 8/27/2024 11:00   anonymous NA    NA
 2     4 8/27/2024 10:59 8/27/2024 11:00   anonymous NA    NA
 3     5 8/27/2024 11:00 8/27/2024 11:00   anonymous NA    NA
 4     6 8/27/2024 10:59 8/27/2024 11:01   anonymous NA    NA
 5     7 8/27/2024 10:59 8/27/2024 11:01   anonymous NA    NA
 6     8 8/27/2024 10:59 8/27/2024 11:01   anonymous NA    NA
 7     9 8/27/2024 10:59 8/27/2024 11:01   anonymous NA    NA
 8    10 8/27/2024 10:59 8/27/2024 11:01   anonymous NA    NA
 9    11 8/27/2024 10:59 8/27/2024 11:01   anonymous NA    NA
10    12 8/27/2024 10:59 8/27/2024 11:01   anonymous NA    NA
# i 176 more rows
# i 22 more variables: `How many pets do you have?` <dbl>,
#   `What kind of pet do you have? (Select all that apply.)` <chr>,
#   `How far (in miles) did you commute to get to class today? (For reference, if you live or
#   `How many minutes have you spent using your phone so far today?` <dbl>,
#   `How old do you think your stats professor is?` <dbl>,
#   `Were you born in the United States?` <chr>, ...
```

Note that I was getting an "Error in nchar(x,"width") : invalid multibyte string, element 1" error to begin with so I found the added argument above on Stack Overflow, which fixed it.

Next, we can see that there are columns here that we won't need and columns with names that are longer than I'd like. There are also columns being treated as numerical which should be logical. So, I'm going to clean this up a bit more and make the tibble more pithily named, so it's easier to type. I

```
 # First, I'll save the names as they are, in case I want to refer to them
  ↪  later in the app.

og_questions<-colnames(student_data)

# Rename columns I'm keeping
studata<-student_data|>
  rename(
    num_pets = 7,
    dist = all_of(9),
    minutes = 10,
    how_old = 11,
    us_born = 12,
    name_ltrs = 13,
    since_movie = all_of(14),
    num_stuffies = 15,
    non_scsu = 16,
    stu_yr = 17,
    rand_num = 18,
    semester = 19,
    acad_yr = 20,
    dog = 21,
    cat = 22,
    reptile = 23,
    fish = 24,
    bird = 25,
    other = 26,
    none = 27)|>

  # Remove unneeded columns
  select(-c(1:6,8,28))|>

  #Change numerical values to 'Yes' or 'No', as appropriate.

  mutate(dog = if_else(dog == 1,"Yes","No"),
         cat = if_else(cat == 1,"Yes","No"),
         reptile = if_else(reptile == 1,"Yes","No"),
         fish = if_else(fish == 1,"Yes","No"),
```

```
        bird = if_else(bird == 1,"Yes","No"),
        other = if_else(other == 1,"Yes","No"),
        none = if_else(none == 1,"No pets","Has pet(s)")) |>

  #Finally, before I start with the summaries, I'll create factors for my
   ↪  qualitative variables and assign the levels, as appropriate.


  mutate(us_bornF = factor(us_born,levels=c("Yes","No")),
        since_movieF = factor(since_movie,levels=c("Less than 7 days","7 -
         ↪  14 days","15 - 30 days","31 - 180 days","More than 180 days or
         ↪  Never")),
        non_scsuF = factor(non_scsu,levels=c("Yes","No")),
        stu_yrF = factor(stu_yr,levels=c("Freshman", "Sophomore", "Junior",
         ↪  "Senior","PSEO/Other")),
        semesterF = factor(semester,levels=c("Fall","Spring")),
        acad_yrF = factor(acad_yr,levels = c("2024/2025",
         ↪  "2025/2026","2026/2027")),
        dogF = factor(dog,levels=c("Yes","No")),
        catF = factor(cat,levels=c("Yes","No")),
        reptileF = factor(reptile,levels=c("Yes","No")),
        fishF = factor(fish,levels=c("Yes","No")),
        birdF = factor(bird,levels=c("Yes","No")),
        otherF = factor(other,levels=c("Yes","No")),
        noneF = factor(none,levels=c("Has pet(s)","No pets"))

        )

og_questions
```

```
 [1] "ID"
 [2] "Start time"
 [3] "Completion time"
 [4] "Email"
 [5] "Name"
 [6] "Last modified time"
 [7] "How many pets do you have?"
 [8] "What kind of pet do you have? (Select all that apply.)"
 [9] "How far (in miles) did you commute to get to class today? (For reference, if you live c
[10] "How many minutes have you spent using your phone so far today?"
[11] "How old do you think your stats professor is?"
[12] "Were you born in the United States?"
```

```
[13] "How many letters are in the name you prefer your friends and family call you?"
[14] "How many long has it been since you've seen a movie in a movie theater?"
[15] "How many stuffed animals do you own?"
[16] "Have you attended an institution besides SCSU after graduating high school (or a schoo
[17] "What year of university are you currently in?"
[18] "Randomly select a whole number between 1 and 10 (inclusive)."
[19] "Semester"
[20] "Academic Year"
[21] "Dog"
[22] "Cat"
[23] "Reptile"
[24] "Fish"
[25] "Bird"
[26] "Other"
[27] "None"
[28] "Primary pet"
```

```
studata
```

```
# A tibble: 186 x 33
   num_pets  dist minutes how_old us_born name_ltrs since_movie     num_stuffies
      <dbl> <dbl>   <dbl>   <dbl> <chr>       <dbl> <chr>                  <dbl>
 1        4   5     60         38 Yes             5 More than 180 ~            3
 2        3  45     65         43 Yes             4 15 - 30 days               3
 3        3  26    120         46 Yes             6 Less than 7 da~            4
 4        1  33      0.11      38 Yes             5 More than 180 ~           20
 5        1  20     48         39 Yes             7 15 - 30 days               2
 6        3   0.1   20         42 Yes             5 31 - 180 days              2
 7        4   0.1   30         38 Yes             3 31 - 180 days              1
 8        3   0.2  138         43 Yes             4 7 - 14 days                2
 9        2   0.1    5         44 Yes             6 15 - 30 days               7
10        3  11     20         30 Yes             6 31 - 180 days              7
# i 176 more rows
# i 25 more variables: non_scsu <chr>, stu_yr <chr>, rand_num <dbl>,
#   semester <chr>, acad_yr <chr>, dog <chr>, cat <chr>, reptile <chr>,
#   fish <chr>, bird <chr>, other <chr>, none <chr>, us_bornF <fct>,
#   since_movieF <fct>, non_scsuF <fct>, stu_yrF <fct>, semesterF <fct>,
#   acad_yrF <fct>, dogF <fct>, catF <fct>, reptileF <fct>, fishF <fct>,
#   birdF <fct>, otherF <fct>, noneF <fct>
```

Looks ready to roll!

Numerical summaries

The first thing I'm going to do is create simple numerical summaries of the numerical data.

```
# First, mean and standard deviation
mean_sum<-apply(X=studata |> select(where(is.numeric)),
                MARGIN = 2,
                FUN = function(x){
                  temp = c(mean(x),sd(x))
                  names(temp)<-c("mean", "sd")
                  temp
                }
)
mean_sum
```

```
     num_pets      dist  minutes  how_old name_ltrs num_stuffies rand_num
mean 2.026882  7.236452 53.10812 39.22043  5.268817      7.50000 6.204301
sd   3.104370 13.031947 53.94623  5.86974  1.786676     10.75891 1.923168
```

Now, the five number summaries:

```
#5 Number summaries
five_num_sum<-apply(X=studata |> select(where(is.numeric)),
                MARGIN = 2,
                FUN = summary
)
five_num_sum
```

```
         num_pets      dist   minutes  how_old name_ltrs num_stuffies  rand_num
Min.     0.000000  0.100000   0.11000 19.00000  2.000000         0.00  1.000000
1st Qu.  0.000000  0.200000  25.00000 35.00000  4.000000         1.00  5.000000
Median   1.000000  0.640000  38.00000 39.00000  5.000000         4.00  7.000000
Mean     2.026882  7.236452  53.10812 39.22043  5.268817         7.50  6.204301
3rd Qu.  3.000000  6.200000  60.00000 43.00000  6.000000         9.75  7.000000
Max.    30.000000 64.000000 400.00000 62.00000 20.000000        65.00 10.000000
```

Categorical summaries

Next, let's look at some qualitative data. First, I'd like to look at how my students break out by class level:

```
studata|>
  group_by(stu_yrF)|>
  summarize(count=n())
```

```
# A tibble: 5 x 2
  stu_yrF    count
  <fct>      <int>
1 Freshman      94
2 Sophomore     41
3 Junior        18
4 Senior         7
5 PSEO/Other    26
```

Being an introductory course, it makes sense that students would be primarily from lower class levels. Note that PSEO a designation for high school students registered at SCSU.

Now, how about looking at students who've attended other institutions, by year.

```
studata|>
  group_by(stu_yrF,non_scsuF)|>
  summarize(count = n())|>
  pivot_wider(names_from = non_scsuF, values_from = count)
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by stu_yrF and non_scsuF.
i Output is grouped by stu_yrF.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(stu_yrF, non_scsuF))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 5 x 3
# Groups:   stu_yrF [5]
  stu_yrF      Yes    No
  <fct>      <int> <int>
1 Freshman      16    78
2 Sophomore     18    23
3 Junior        10     8
4 Senior         5     2
5 PSEO/Other     1    25
```

Again, these results make sense. Lower level students (Freshman and Sophomore) are much less likely to have transferred.

Now I'm curious about whether there's a difference in class level versus Semester. My guess is that more freshman focus on STAT 103 (a general education course) during the fall.

```
studata|>
  group_by(stu_yrF,semesterF)|>
  summarize(count = n())|>
  pivot_wider(names_from = semesterF, values_from = count)
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by stu_yrF and semesterF.
i Output is grouped by stu_yrF.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(stu_yrF, semesterF))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 5 x 3
# Groups:   stu_yrF [5]
  stu_yrF     Fall Spring
  <fct>      <int>  <int>
1 Freshman      70     24
2 Sophomore     33      8
3 Junior        14      4
4 Senior         4      3
5 PSEO/Other    24      2
```

Interestingly, Fall is more popular in roughly the same proportion across the board, except for PSEO, which is not something I have a good explanation for.

Numerical and categorical Summaries

Next, lets look at some numerical summaries broken down at levels for categorical data.

First up, I'm interested in the mean number of stuffed animals the students own, separated by whether they have pets.

```
studata|>
  group_by(noneF)|>
  summarize(mean_stuffies = mean(num_stuffies))
```

```
# A tibble: 2 x 2
  noneF      mean_stuffies
  <fct>              <dbl>
1 Has pet(s)          9.03
2 No pets             3.85
```

I'm a little surprised, actually, that the difference is so stark here.

Next up, I'm curious about how students have judged my age over the last few years.

```
studata|>
  group_by(acad_yrF)|>
  summarize(mean_guess = mean(how_old))
```

```
# A tibble: 3 x 2
  acad_yrF  mean_guess
  <fct>          <dbl>
1 2024/2025       39.5
2 2025/2026       39.3
3 2026/2027       38.6
```

Clearly, my exposure to Benjamin Button is having the desired effect. Sadly, the decrease of more than a year between Fall 2025 and Fall 2026 is not directly observable in the table. Let's see if we can make it so.

```
studata|>
  group_by(semesterF,acad_yrF)|>
  summarize(mean_guess = mean(how_old))|>
  pivot_wider(names_from = acad_yrF, values_from = mean_guess)
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by semesterF and acad_yrF.
i Output is grouped by semesterF.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(semesterF, acad_yrF))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 2 x 4
# Groups:   semesterF [2]
  semesterF `2024/2025` `2025/2026` `2026/2027`
```

```
  <fct>           <dbl>       <dbl>       <dbl>
1 Fall             39.4        40.1        38.6
2 Spring           39.6        37.5        NA
```

There we go! We don't need to consider Spring 2026 to Fall 2026.

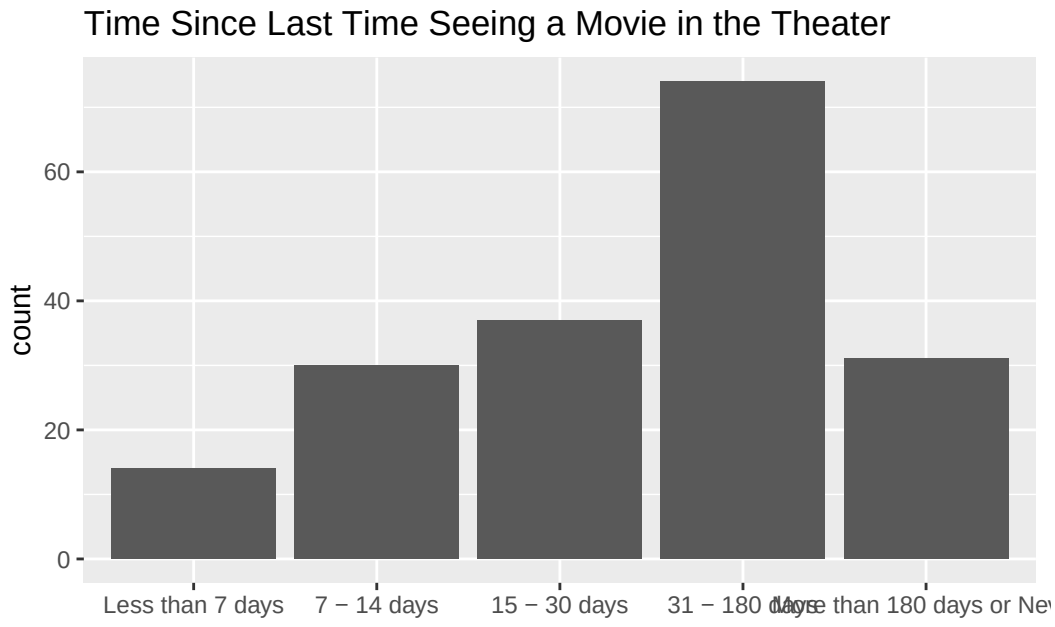
Now, for some...

Plots

To keep things simple to start, I'm going to make a nice bar graph of how long it's been since the students have seen a movie.

```
movie_bar<-ggplot(data=studata,aes(x=since_movieF))+geom_bar()+labs(
    title = "Time Since Last Time Seeing a Movie in the Theater",
    x = "")

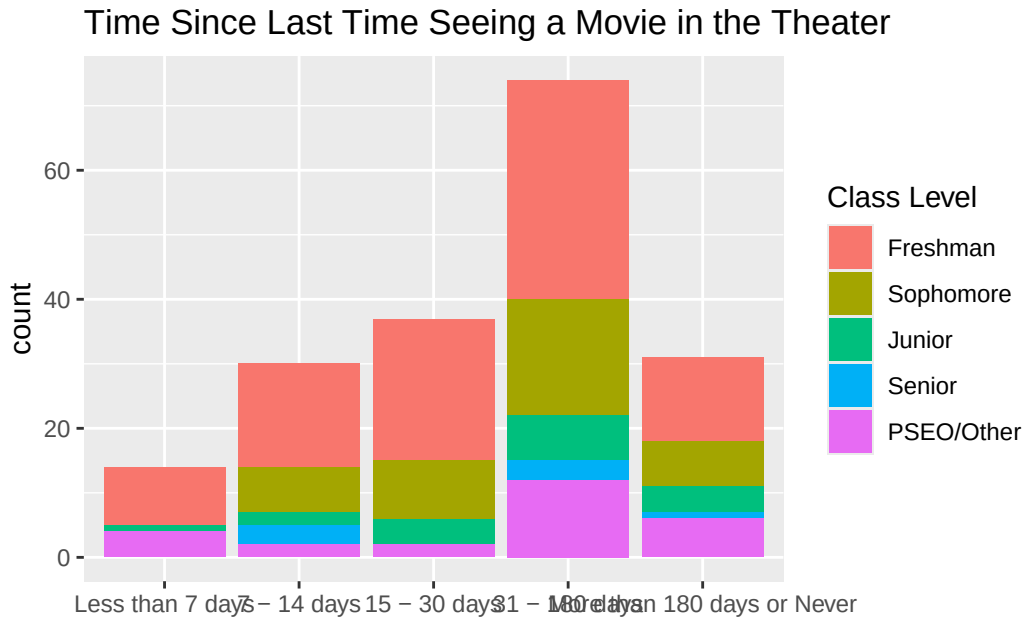
movie_bar
```



We can make this a little more interesting. First, I'm going to see if I can make break the bars down by class level.

```
movie_bar2<-ggplot(data=studata,aes(x = since_movieF, fill =
 ↪  stu_yrF))+geom_bar()+labs(
    title = "Time Since Last Time Seeing a Movie in the Theater",
    x = "") + scale_fill_discrete("Class Level")

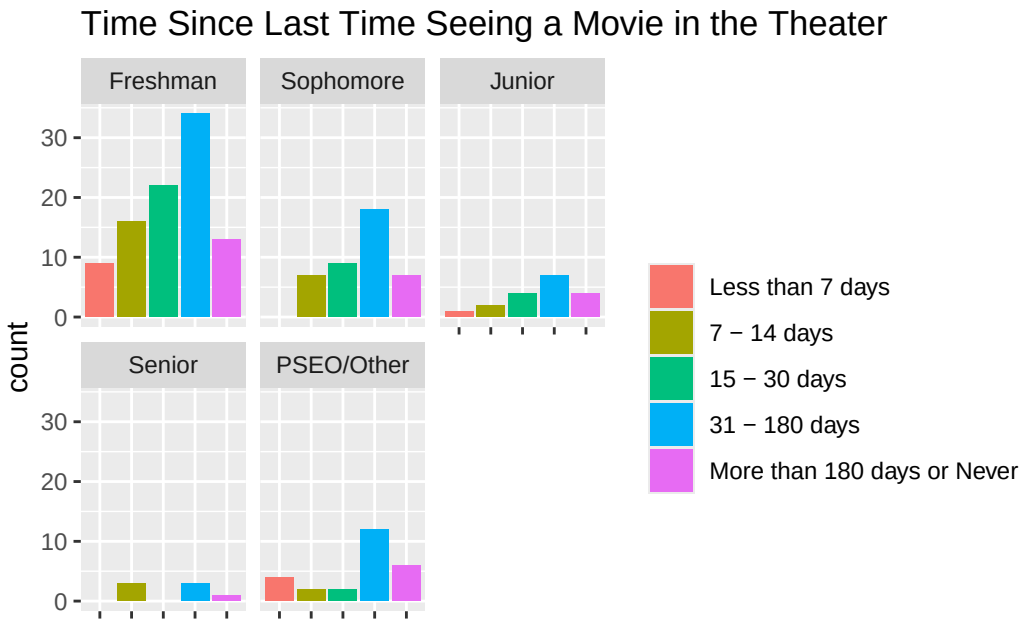
movie_bar2
```



That's informative, I guess, if more than a little busy. Let's add the class level as a facet instead.

```
movie_bar3<-ggplot(data=studata,aes(x = since_movieF, fill =
 ↪  since_movieF))+geom_bar()+labs(
    title = "Time Since Last Time Seeing a Movie in the Theater",
    x = "") + facet_wrap(~ stu_yrF) + scale_fill_discrete("") +
     ↪  theme(axis.text.x = element_blank())

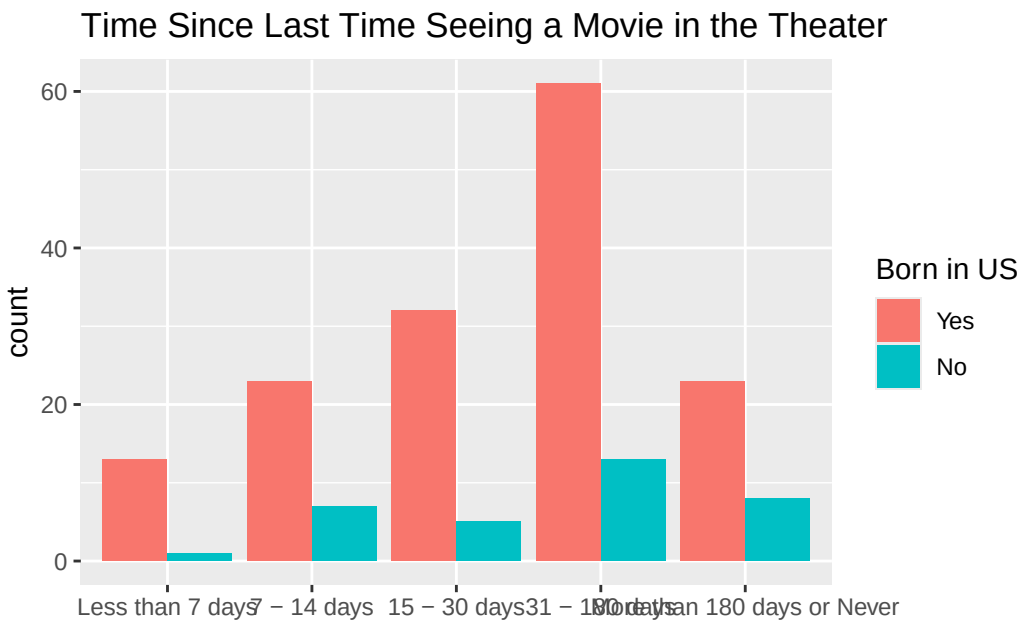
movie_bar3
```



Next, I'll add a dodge so I can see if there's a difference when looking at students not born in the United States.

```
movie_bar4<-ggplot(data=studata,aes(x = since_movieF, fill =
 ↪  us_bornF))+geom_bar(position = "dodge")+labs(
    title = "Time Since Last Time Seeing a Movie in the Theater",
    x = "") + scale_fill_discrete("Born in US")

movie_bar4
```

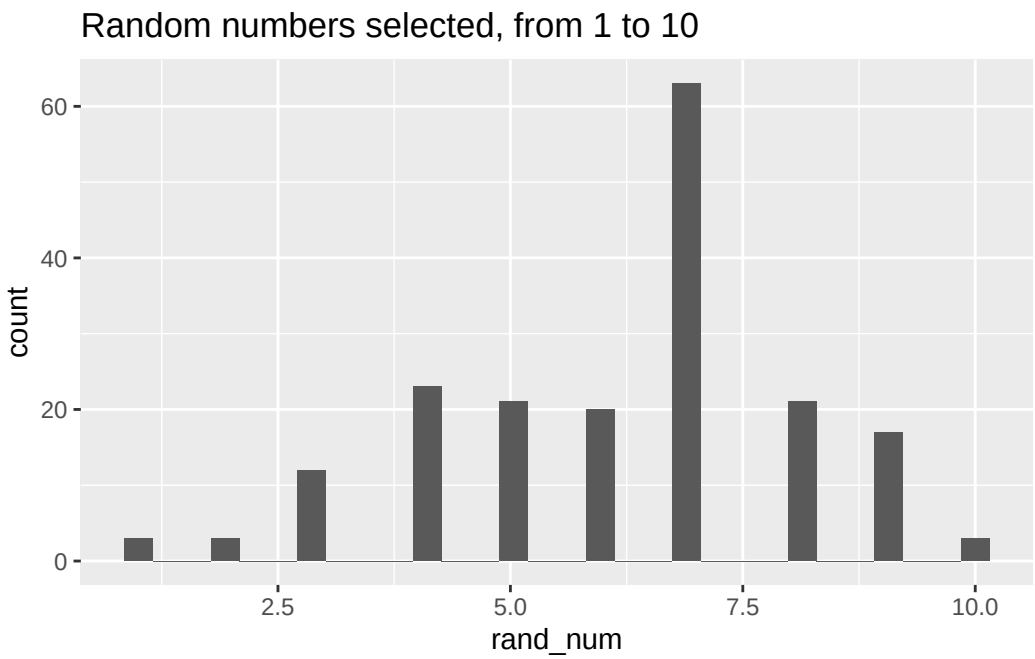


Other than the raw numbers being different, I don't see a substantial difference in moving going behavior here.

Next, let's look at a few numerical variables. One I find very interesting is what happens when I ask students to enter a random number. If you haven't seen it, you might find the video that inspired the question interesting. Anyway, here's the histogram of the students' "random" numbers.

```
hist<-ggplot(data = studata, aes(rand_num))+geom_histogram()+labs(title =
 ↪  "Random numbers selected, from 1 to 10")
hist
```

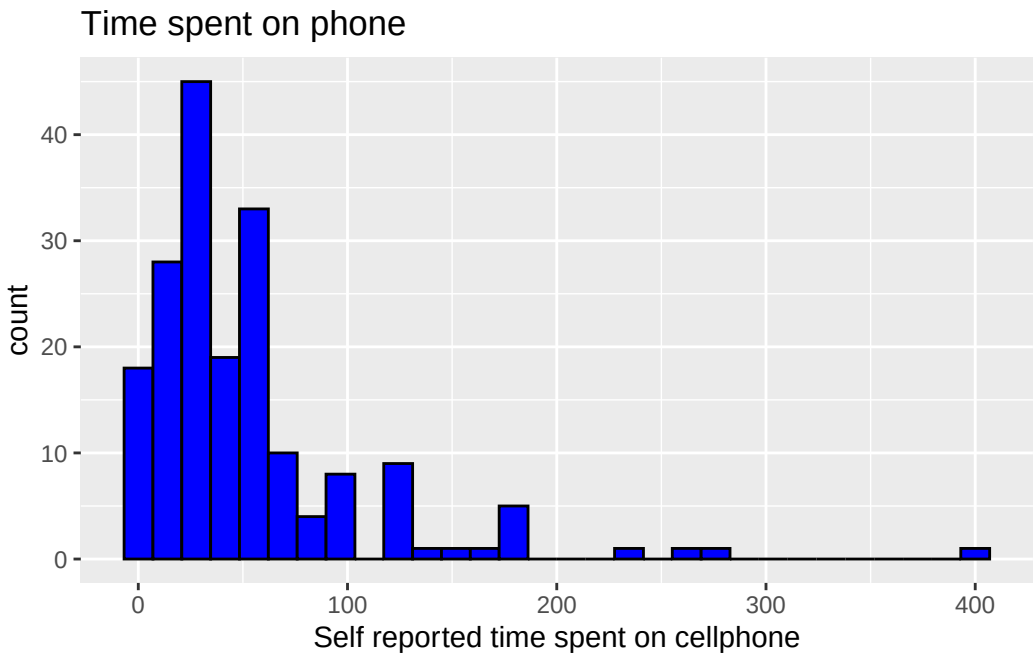
```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



I get that one could argue that this shouldn't be considered numeric because nothing's being measured, but I do think students use some "numberness" in making their selections. Ok, fine, you're not convinced. Here's a histogram for the number of minutes students reporting spending on their phone:

```
hist2<-ggplot(data = studata,
 ↪  aes(minutes))+geom_histogram(fill="blue",color="black",linetype="solid")+labs(title
 ↪  = "Time spent on phone", x = "Self reported time spent on cellphone")
hist2
```

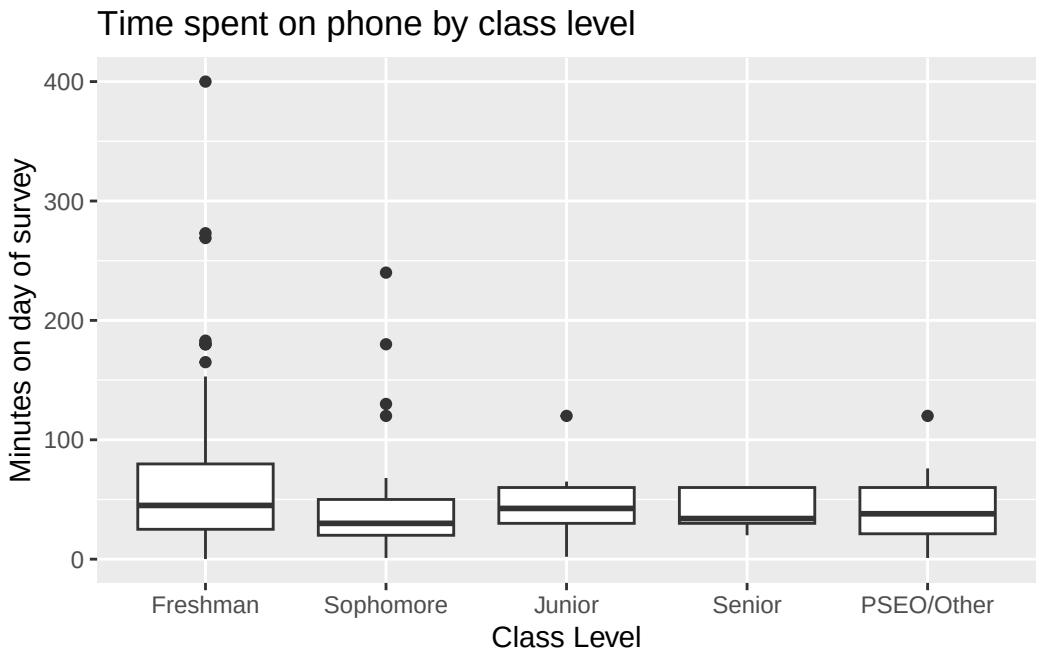
```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Next, I'd like to look at the relationship between students' time spent on their phones and their class level. I'm wondering if being a younger student corresponds in more phone time.

```
box<-ggplot(data = studata, aes(x = stu_yrF, y = minutes)) + geom_boxplot() +
 ↪  labs(title = "Time spent on phone by class level", x = "Class Level", y =
 ↪  "Minutes on day of survey")

box
```

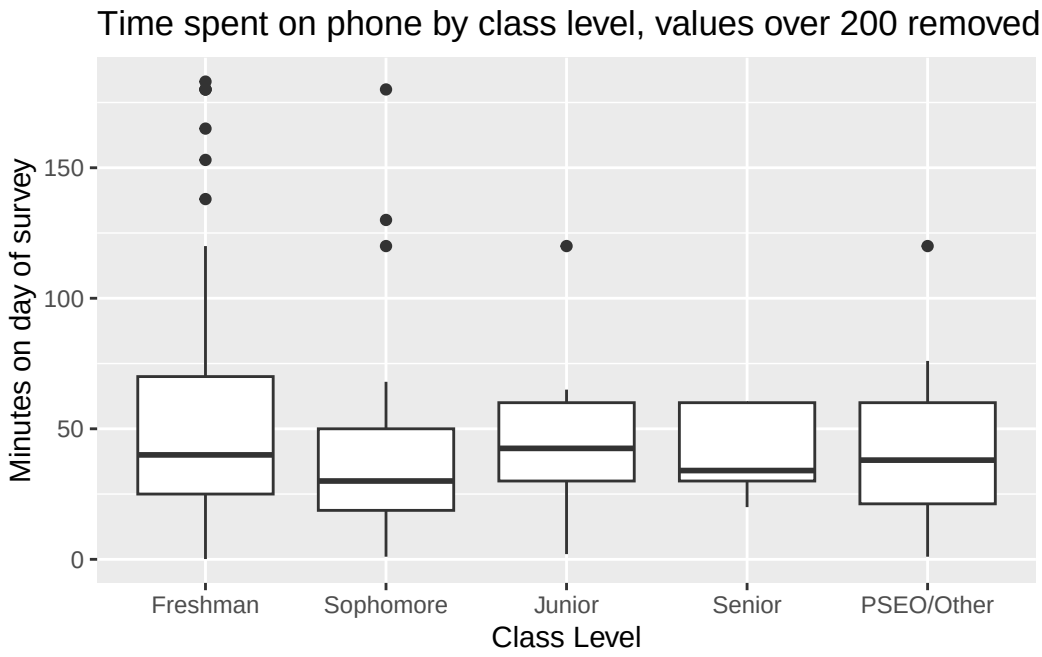


This worked, but the outliers in the freshman class are making it hard to see the other boxplots too well.

```
#Remove data values over 200 minutes
temp_data<-studata|>filter(minutes<=200)

box2<-ggplot(data = temp_data, aes(x = stu_yrF, y = minutes)) +
 ↪  geom_boxplot() + labs(title = "Time spent on phone by class level, values
 ↪  over 200 removed", x = "Class Level", y = "Minutes on day of survey")

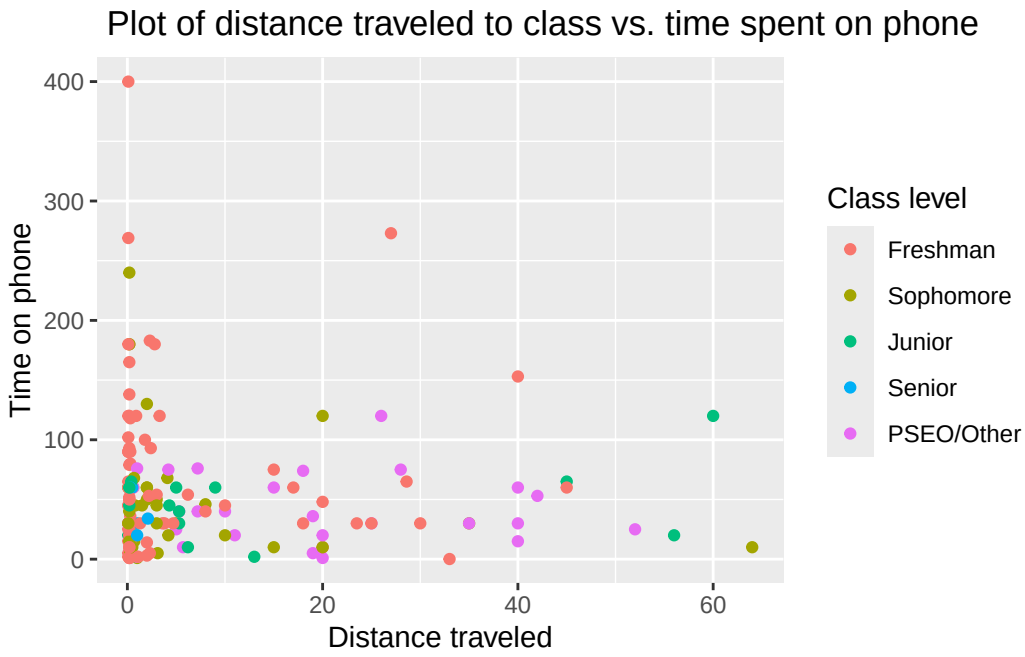
box2
```



Next, I'd like to look at the relationship between distance traveled to class and minutes spent on phone, in a scatter plot.

```
scat<-ggplot(studata, aes(x = dist, y = minutes, color = stu_yrF)) +
 ↪  geom_point() + labs(title = " Plot of distance traveled to class vs. time
 ↪  spent on phone", x = "Distance traveled" , y = "Time on phone", color =
 ↪  "Class level")

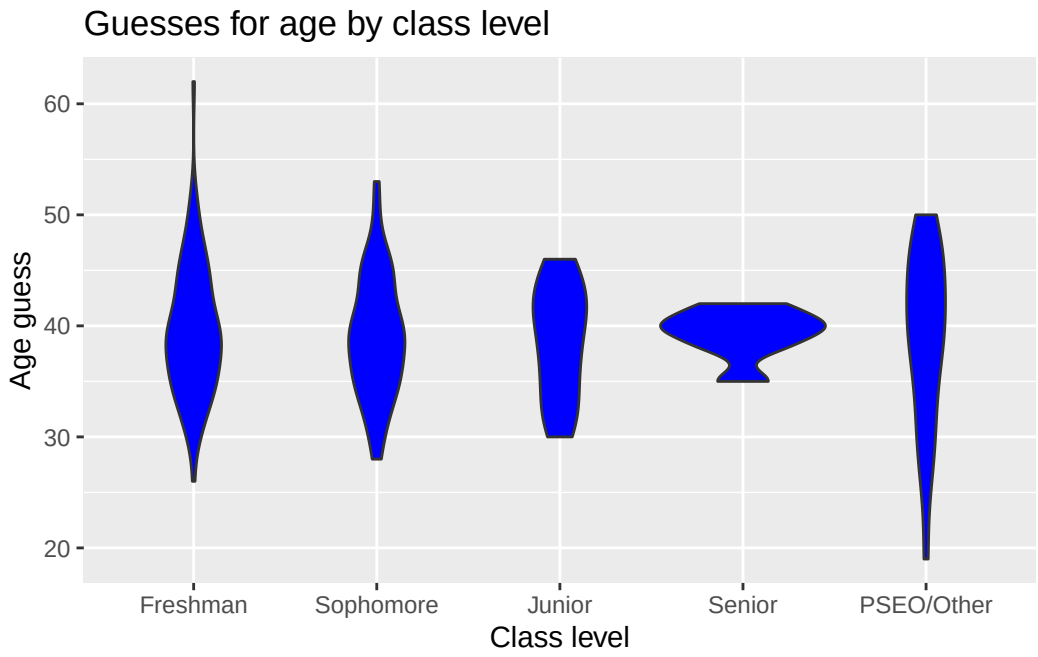
scat
```



Finally, I'd like to look at a violin plot (I don't think we did one of those in class) of students' guesses for my age, based on their class level. I'm curious if older students are better guessers.

```
age_vi<-ggplot(studata, aes(x = stu_yrF, y = how_old)) + geom_violin(scale =
 ↪  "area", fill = "blue") + labs(title = "Guesses for age by class level", x
 ↪  = "Class level" , y = "Age guess")

age_vi
```



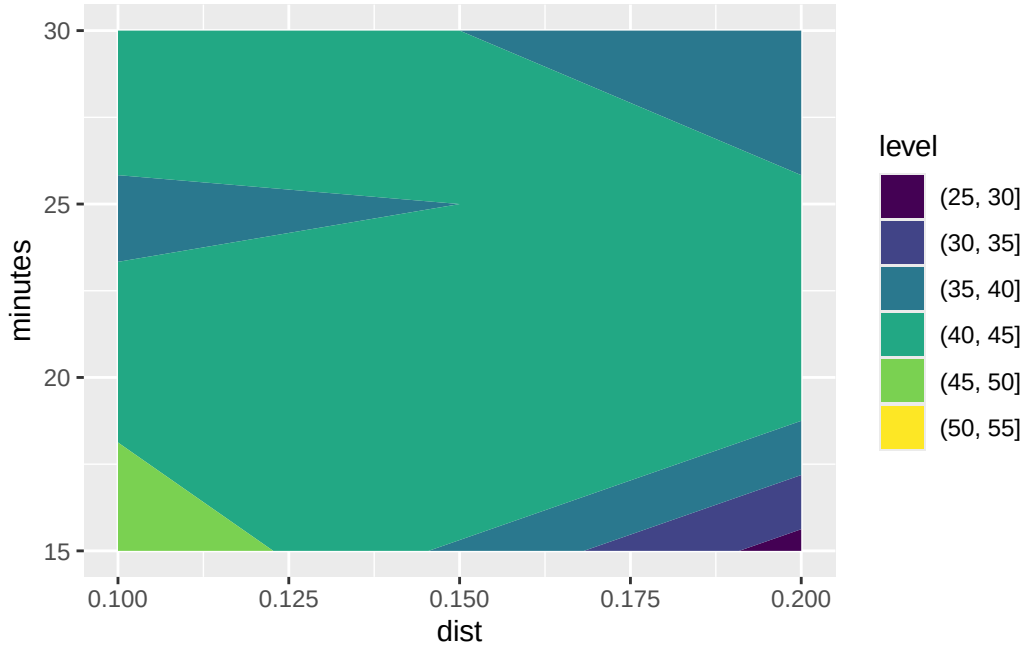
It sure seems like the younger the students are, the more all-over-the-place their guesses become.

Finally, I'd like to give making a heat map a try, even though I don't think it'll be particularly meaningful here.

```
heat<-ggplot(studata, aes(x = dist, y = minutes)) + geom_contour_filled(aes(z
 ↪  = how_old))

heat
```

```
Warning: Contour data has duplicated x, y coordinates.
i 44 duplicated rows have been dropped.
```



As expected, this isn't terribly useful but it's here! It's interesting that there were 44 records dropped. I wouldn't have thought there were that many students who reported the same distance and minutes. Looking at the data, though, it's because many students rounded to either 30 or 60 minutes and students who live on campus often entered 0.1 or 0.2 miles, creating a lot of duplication for these variables.